

RFI — IMU leg wearable for equine monitoring (BSV EquiCare)

To: IMU / wearable-sensor vendor

From: Bharat Sports Venture — EquiCare engineering

Covers client monitoring points: 1 (steps / locomotion), 5 (activity), 6 (rest & lying), 7 (lameness / limb-favouring)

0. How to reply — please read first

Please answer all of this in ONE consolidated reply. We are evaluating several suppliers in parallel on a fixed timeline and will not be sending rounds of follow-up questions. In practice that means:

1. **Answer every numbered question**, in the reply slots provided. If something is not supported, **write that plainly** — "not supported" is a perfectly good answer and costs you nothing. A blank or skipped item, however, we have to record as *not supported*, because we cannot tell the difference.
2. **Attach the documents and files listed in the checklist at the end.** Please send them with this reply, not "on request" — a claim we cannot verify counts as unverified.
3. **Send the actual artifact, not a description of it.** Our single worst experience with a previous supplier: they confirmed "Linux SDK — yes", and the package that arrived was x86-64 only, with no ARM64 build available at all. That cost both sides weeks. So where we ask for a binary, driver, register map or protocol spec, please attach the real file (or a download link) — or state clearly that it does not exist.
4. If a question does not apply to your product, write **"N/A"** and one line saying why.

We would rather receive an honest reply with several "not supported" answers than an optimistic one that unravels at integration.

1. What we are building

A 24/7 monitoring system for stabled horses in India. Each stall has sensors feeding an **on-site edge computer** (NVIDIA Jetson); our software turns raw data into health and behaviour insights. We need a **leg-mounted IMU tag** (cannon bone) streaming **raw motion data** to that edge box.

All analytics — step counting, gait analysis, lameness, lying detection — are done by us. We are not buying your algorithms or dashboard. We need **raw, timestamped, per-axis data** and a documented way to read it on our own hardware.

2. Hard requirements — please confirm these first

Pass/fail for us. If any cannot be met, please say so plainly rather than leaving it blank — it saves us both time.

#	Requirement	Why
H1	Linux ARM64 (aarch64) support — NVIDIA Jetson. Any SDK/driver must be <i>built and tested</i> for aarch64, not only x86-64.	Our edge box is a Jetson. A previous vendor shipped x86-only binaries and could not rebuild — it blocked us for weeks. If your tag speaks an open protocol (BLE GATT / serial / MQTT), say so and this is a non-issue.
H2	Raw per-axis time-series accessible — not only step counts or vendor-computed summaries.	Our gait/lameness models need the waveform. A closed product that only returns "activity score" cannot work for us.
H3	No mandatory vendor cloud. Data must reach our edge box over the local network.	India DPDP-2023 residency, and barns with unreliable internet.
H4	India WPC/ETA radio approval + BIS.	Legal deployment requirement.

REPLY (H1-H4)

3. Sensing

Please answer with **specific values / part numbers**, not "yes — supported".

- Sensor part number(s) + datasheet.** Is it **6-axis** (3-axis accel + 3-axis gyro) or **9-axis** (with magnetometer)? *6-axis is our minimum; 9-axis preferred.*

REPLY

- Accelerometer full-scale range**, and whether it is selectable ($\pm 2/4/8/16$ g). We require $\geq \pm 16$ g — a horse kicking or striking the stall wall will clip lower ranges and corrupt the very data we need.

REPLY

- Gyroscope full-scale range** (selectable ± 250 – 2000 °/s?). We require $\geq \pm 2000$ °/s — no clipping at trot, canter or kick.

REPLY

4. **Configurable output data rate (ODR).** We require ≥ 100 Hz, **target 100–200 Hz**. State supported rates, whether it is configurable **over the air**, and the achievable rate *while streaming raw*.

REPLY

5. **Noise density / RMS noise** for accel and gyro at our target ODR. State any on-chip filtering and whether it can be bypassed.

REPLY

6. **Gyro bias drift with temperature** (Indian barns reach 45–50 °C), and whether the tag can be **recalibrated in the field without removing it** from the horse.

REPLY

7. **On-firmware activity classification** (standing / lying / walking / trotting / restless) — do you provide it, is it **equine-validated or generic**, and **can it be disabled**? We classify on our edge, so we need raw data regardless; we do not want firmware quietly pre-filtering it.

REPLY

8. **Step counting** — equine-validated or adapted from human/cattle? State error at walk and at trot.

REPLY

9. **Per-reading quality / confidence flag** — any validity, saturation or signal-quality indicator per sample or per batch, that we can carry into our data model?

REPLY

4. Data access and integration

10. **How does raw data leave the tag?** State the transport (BLE GATT, Wi-Fi, proprietary radio → gateway, USB, serial) and **attach the protocol/API documentation or register map with this reply**, not later.

REPLY

11. **Streaming vs. batch.** Can the tag **stream continuously** at ≥ 100 Hz, or only upload periodic batches? If batched, state interval, payload format and throughput.

REPLY

12. **Documented output schema and units** — exact field names and units for accel (g or m/s^2) and gyro ($^\circ/\text{s}$ or rad/s). Confirm it can be mapped to our contract: `{horseId, metric, value, unit, ts, source, confidence}`. Is the schema versioned? Sample payload welcome.

REPLY

13. **BLE specifics** (if BLE): version (4.2 / 5.x), **Coded-PHY / Long-Range** support, TX power, and **realistic range through masonry and steel barn structure** — not open-field figures.

REPLY

14. **Gateway** (if required): model, **how many tags per gateway at our raw sample rate**, uplink (Ethernet/PoE preferred, Wi-Fi, 4G), and confirmation of a **local-only path to our edge box**.

REPLY

15. **Any vendor cloud in the data path** — and if so, where are those servers located? Can the entire path be **edge-only / on-premise with no data leaving India**?

REPLY

5. Time synchronisation

16. **Are samples timestamped on the tag** or on receipt? State the **clock-sync method (PTP preferred, NTP acceptable)**, worst-case **drift/skew between tags over 24 h**, and whether the gateway can discipline tag clocks. *We must align IMU data with video frames, thermal ROI temperatures and audio on one timeline — we are targeting tens of milliseconds.*

REPLY

6. Resilience and power

17. **On-tag buffering** — we require **≥24 h** with **automatic backfill using the original timestamps** and acknowledged delivery (barns lose power and connectivity). State capacity in hours **at 100 Hz raw**.

REPLY

18. **Battery life** at each mode (lying-detection only vs 100–200 Hz raw streaming); chemistry/mAh; rechargeable or replaceable; charge time and cycle life.

REPLY

19. **Charging operations for a whole yard** — multi-bay dock, hot-swap spares, how data gaps during a swap are handled, and how a replacement tag is **re-bound to the same horse**.

REPLY

7. Fit, welfare and environment

20. **IP rating** — **we require IP67 or better**, validated against wash-down, mud, urine/manure and sweat. State operating temperature and humidity range (we need 45 °C+ and monsoon).

REPLY

21. **Attachment** to the cannon bone (strap / boot pocket), size range **pony** → **draft**, and expected strap service life.

REPLY

22. **Welfare** — tag weight and dimensions (*target* ≤ 50 g), and any 24/7 wear assessment showing no rub sores or skin damage. Please share equine field-trial or welfare data.

REPLY

23. **Loss prevention** — **tamper/detach alert**, and your observed **tag-loss rate per tag-year** in field deployments.

REPLY

8. Identity and fleet management

24. **Globally unique immutable hardware ID** present in every record, and how we **bind tag** → **horseld** via API (queryable mapping).

REPLY

25. **Fleet management** — firmware OTA, battery/health monitoring, remote configuration, and tenant isolation if you host anything.

REPLY

9. Evaluation

We would like **3–5 evaluation tags + 1 gateway + charging dock**, together with the **SDK/driver, protocol documentation and sample code** for Linux/ARM64, so we can validate on our Jetson bench before committing to a design.

REPLY (AVAILABILITY + LEAD TIME FOR EVALUATION HARDWARE)

10. Optional — equine validation data

If you hold any **equine** reference or validation data (step counts, lying time, gait asymmetry against a gold standard), please share it. Most tags are validated on cattle or humans; equine gait differs, and this would materially reduce our modelling risk.

REPLY

Response checklist

Please confirm each item is attached or answered, so we can evaluate in a single pass:

	Item
☐	Every numbered question answered (or marked "not supported" / "N/A")
☐	All hard requirements (H-numbers) explicitly confirmed or declined
☐	Sensor / component datasheets
☐	Protocol document, register map or API reference — the actual file
☐	ARM64 (aarch64) build attached or download link — <i>or</i> a plain statement that none exists
☐	Sample data payload showing real field names and units
☐	Sample code, if you provide any
☐	Evaluation-unit availability and lead time
☐	A named technical contact we can reach directly with any single clarification

If anything on this list genuinely cannot be shared before an order, please say which item and why, so we can factor it in rather than assume the worst.

Commercial terms are not part of this RFI — we will discuss those separately once the technical fit is confirmed.